

NAME: Nageswara Rao | Email: k.nrao8844@gmail.com | Ph: +1 774 231 8844
Sr. Data Engineer/Big Data Engineer

BACKGROUND SUMMARY:

- Senior Data Engineer with 11 years of experience in designing, developing, and optimizing large-scale data pipelines, data platforms, and real-time streaming solutions across industries such as **banking, telecommunications, and financial services**.
- Expertise in building end-to-end data pipelines using a comprehensive tech stack, including Hadoop ecosystem (**Spark, Hive, HBase, Sqoop, Oozie, Flume, Zookeeper**), streaming frameworks (Kafka, Spark Streaming), and cloud platforms (**AWS, Azure**).
- Extensive cloud migration experience, leading the transition of on-premises databases to AWS S3, Redshift, and Azure Data Lake, resulting in 60% improvement in storage scalability and significant cost optimization.
- Led **cross-functional teams**, mentoring junior engineers and guiding data engineering best practices. Played a key role in **sprint planning, backlog grooming**, and code reviews to ensure high-performance and scalable data solutions.
- Skilled in developing high-performance ETL processes using **Python, Scala, and PySpark**, processing both structured and unstructured data from diverse sources such as **SQL Server, Oracle, Netezza, DB2, and streaming feeds**.
- Proven track record of reducing data processing times by 40% through optimized Spark jobs, effective caching, partitioning strategies, and real-time streaming pipelines.
- Expertise in **PL/SQL, Python, and ETL tools**, with a strong background in **cloud platforms (AWS, Azure), data modeling, and performance tuning**.
- Experience with data modeling (Star & Snowflake schemas), data warehouse development, and performance tuning using relational databases like SQL Server, Oracle, Teradata and NoSQL databases like HBase, Cassandra, and MongoDB.
- Hands-on in configuring and administering Hadoop clusters (**Cloudera and Hortonworks**), ensuring high availability, fault tolerance, and optimal resource management.
- Developed and maintained automated CI/CD pipelines for data workflows using Jenkins, Azure DevOps, and GitHub Actions, reducing manual deployment efforts by 40%.
- Strong experience in batch processing with MapReduce and Spark for analytical processing, data aggregation, cleansing, and transformation, handling terabytes of data efficiently.
- Expertise in scripting and automation using **Shell scripting, Python, and Perl, automating ETL tasks**, system monitoring, and data quality checks.
- Developed and optimized Spark SQL queries for ad-hoc analysis and batch transformations, improving query performance by up to 50% using indexing, partitioning, and caching techniques.
- Hands-on experience designing and implementing role-based access controls (RBAC) and secure data access policies in Azure Active Directory and AWS IAM.
- Collaborated with **cross-functional teams** (data scientists, analysts, and business stakeholders) to understand business requirements and design data pipelines that support advanced analytics and reporting.
- Experience with containerized deployments using **Docker and Kubernetes**, enhancing pipeline portability and scalability.
- Hands-on with **Azure Databricks** for data transformation and analytics, leveraging Delta Lake for efficient data lake management.
- Strong troubleshooting and debugging skills across Hadoop ecosystem tools, Spark performance tuning, and cloud resource optimization.
- Experience working in **Agile/Scrum** environments, contributing to sprint planning, backlog grooming, and delivering iterative improvements to data pipelines.

TECHNICAL SKILLS:

Bigdata/Hadoop Technologies	MapReduce, Spark, SQL, Azure, Spark Streaming, Kafka, PySpark, Pig, Hive, HBase, Flume, Yarn, Oozie, Zookeeper, Hue, Ambari Server
Languages	HTML5, DHTML, WSDL, CSS3, C, C++, XML, R/R Studio, SAS Enterprise Guide, SAS, R (Caret, Weka) Perl, MATLAB, Mathematica, FORTRAN, DTD, Schemas, Json, Ajax, Java, Scala, Python (NumPy, SciPy, Pandas, Genism, Keras), Java Script, Shell Scripting
NO SQL Databases	Cassandra, HBase, MongoDB, MariaDB
Web Design Tools	HTML, CSS, JavaScript, JSP, jQuery, XML
Development Tools	Microsoft SQL Studio, IntelliJ, Azure Databricks, Eclipse, NetBeans.
Cloud	AWS (S3, CloudWatch, Athena, RedShift, EMR, EC2, DynamoDB, IAM, Secret Manager, S3, Lambda, CloudWatch, Messaging Queue (SNS & SQS)), Azure (Azure Data Factory, Azure Blob, Azure Databricks, ADF, Blob Storage), GCP – Big Query, Autoscaling,
Development Methodologies	Agile/Scrum, UML, Design Patterns, Waterfall
Build Tools	PowerBI, Tableau, Jenkins, Toad, SQL Loader, PostgreSQL, Talend, Maven, ANT, RTC, RSA, Control-M, Oziee, Hue, SOAP UI
Reporting Tools	MS Office (Word/Excel/Power Point/ Visio/Outlook), Crystal reports XI, SSRS, Cognos.
Databases	Microsoft SQL Server, MySQL, Oracle, DB2, PL/SQL, Teradata, Netezza
Operating Systems	All versions of Windows, UNIX, LINUX, Macintosh HD, Sun Solaris

WORK EXPERIENCE:

Client: Fifth Third Bank, Cincinnati

Sept 2023 - Present

Role: Senior Big Data Engineer

Responsibilities:

- Designed and implemented scalable ETL pipelines for real-time and batch data processing, enabling efficient data collection, transformation, and storage.
- Developed PL/SQL procedures, functions, and triggers to streamline database operations and improve processing efficiency.
- Built robust data ingestion frameworks to collect, cleanse, and transform data from multiple structured and unstructured sources.
- Optimized data workflows by implementing advanced partitioning, indexing, and caching techniques to improve performance and reduce processing time by 40%.
- Ensured data security, compliance, and governance by implementing encryption techniques and access control measures across cloud environments such as AWS and Azure.
- Developed Python-based automation scripts for ETL workflows, eliminating manual intervention and increasing efficiency.
- Led root cause analysis for performance issues in SQL queries, optimizing queries for large-scale SQL and NoSQL databases to enhance system stability.
- Partnered with data scientists and analysts to design data models that support predictive analytics and decision-making initiatives.

Environment: Big Data, Hadoop, Oracle, PL/SQL, Scala, Spark-SQL, PySpark, Python, Kafka, Sas, SQL, Mdm, Oozie, Ssis, T-SQL, Etl, Hdfs, Cosmos, Pig, Sqoop, Ms Access.

Client: Charter Communications, Denver, Colorado

June 2021 - Aug 2023

Role: Senior Big Data Engineer

Responsibilities:

- Designed and developed Azure Data Factory (ADF) pipelines using Linked Services, Datasets, and Pipelines for ETL workflows across Azure SQL, Blob Storage, and Synapse Analytics.
- Developed and optimized Spark applications (Scala/PySpark) for real-time data processing using Spark Streaming and Kafka.
- Created MapReduce jobs for log file analysis stored in Hadoop clusters, improving query performance.
- Integrated structured and unstructured data using Hive and Pig for business insights.
- Migrated multiple on-premises solutions to Azure Cloud, including IaaS infrastructure, network integration, and cloud storage solutions.
- Implemented distributed messaging queues using Apache Kafka and Zookeeper for seamless data integration with Cassandra and Elasticsearch.
- Developed ADF JSON scripts for deploying pipelines that process data using Cosmos Activity.
- Optimized Spark jobs using RDD caching, partitioning, and broadcast joins, improving efficiency by 40%.
- Managed data movement between HDFS and RDBMS using Apache Sqoop for batch processing.
- Built ETL workflows using Talend, Informatica, and SQL-based transformations for seamless data integration.
- Automated ETL workflows using Sqoop and Oozie, ensuring efficient data ingestion from relational databases to Hadoop.
- Utilized Azure Databricks & DataFrames for data transformation, working with Parquet, SQL tables, and Hive external tables.
- Developed complex SQL queries, stored procedures, and functions for PostgreSQL, Oracle, and MySQL to support Big Data processing and transactional databases.
- Enforced security and compliance by implementing Role-Based Access Control (RBAC) and security policies in Azure Active Directory.
- Designed and implemented Kafka-HBase and Kafka-Elasticsearch pipelines for real-time event-driven data storage and retrieval.
- Developed Spark-Kafka streaming pipelines for ingesting high-velocity Kafka data into Spark for real-time analytics.
- Used Spark Streaming to capture and store real-time data into HDFS, HBase, and Cassandra.
- Built workflow scheduling pipelines using Apache Oozie, Zookeeper, Airflow, and Apache NiFi.
- Deployed containerized Big Data applications using Docker & Kubernetes for improved scalability and performance.
- Benchmarked and optimized Hadoop/HBase clusters to enhance performance and scalability.
- Developed Python scripts leveraging pandas, NumPy, scikit-learn, and Plotly for data visualization and machine learning.
- Designed and implemented PL/SQL program units, triggers, stored procedures, and functions to enforce business rules in enterprise databases.
- Automated SQL query execution using Python scripts, enabling parallel data loading into target tables.
- Created Hive tables, optimized queries, and developed Spark DataFrame APIs in Scala for advanced data analysis and transformation.
- Utilized AWS EMR to run Apache Hadoop, CDH, and MapR distros on EC2 for elastic computing.
- Developed Scala-based log producers to track incremental application logs and stream them to Kafka and Zookeeper-based platforms.
- Built log ingestion and transformation pipelines using Kafka, Spark Streaming, and HDFS for high-speed analytics.

Environment: HDFS, Hive, Spark, Linux, Kafka, python, Azure, Stone branch, Cloudera, Oracle11g/10g, PL/SQL, Unix, Json and Parquet File systems.

Client: Edward Jones, St. Louis, MO

Mar 2019 - May 2021

Role: Big Data Engineer

Responsibilities:

- Involved in HBASE setup and storing data into HBASE, which will be used for further analysis.
- Application development using Hadoop Ecosystems such as Spark, Kafka, HDFS, HIVE, Oozie and Sqoop.
- Experience in developing customized UDF's in Python to extend Hive and Pig Latin functionality.
- Involved in SQOOP implementation which helps in loading data from various RDBMS sources to Hadoop systems and vice versa.
- Involved in Analysing system failures, identifying root causes, and recommended course of actions, Documented the systems processes and procedures for future references.
- Designed AWS Cloud Formation templates to create VPC, subnets, NAT to ensure successful deployment of Web applications and database templates.
- Worked on a python script to extract data from Netezza databases and transfer it to AWSS3
- Developed and designed automate process using shell scripting for data movement.
- Developed workflow in Oozie to automate the tasks of loading the data into Nifi and pre-processing with Pig.
- Application development using Hadoop Ecosystems such as Spark, Kafka, HDFS, HIVE, Oozie and Sqoop.
- Built S3 buckets and managed policies for S3 buckets and used S3 bucket and Glacier for storage and backup on AWS
- Performed data manipulation on extracted data using Python Pandas.
- Designed AWS Cloud Formation templates to create VPC, subnets, NAT to ensure successful deployment of Web applications and database templates.
- Created SAS ODS reports using SAS EG, SAS SQL, and OLAP Cubes.
- Implemented the SparkScala code for Data Validation in Hive
- Designed and implemented big data ingestion pipelines to ingest multi PB data from various data source using Kafka, Spark streaming including data quality checks, transformation, and stored as efficient storage formats like parquet.
- Implemented the automated workflows for all the jobs using the Oozie and shell script.
- Implemented python codebase for branch management over Kafka features.
- Utilized SQOOP, Kafka, Flume and Hadoop File system APIs for implementing data ingestion pipelines
- Worked on results from Kafka server output successfully.
- Work with subject matter experts and project team to identify, define, collate, document and communicate the data migration requirements.
- Designed and implemented big data ingestion pipelines to ingest multi PB data from various data source using Kafka, Spark streaming including data quality checks, transformation, and stored as efficient storage formats like parquet.
- Extensively worked with Partitions, Dynamic Partitioning, bucketing tables in Hive, designed both Managed and External tables, also worked on optimization of Hive queries.
- Designed Oozie workflows for job scheduling and batch processing.
- In depth knowledge on import/export of data from Databases using Sqoop.

Environment: Hadoop, Kafka, Spark, Sqoop, Hdfs, Map reduce, Scala, Mango DB, HBase, Jenkins, AWS, Lambda, Talend, SAS, Unix, Oracle

Client: UBS, weehawken, NJ

January 2017 - Feb 2019

Role: Data Engineer

Responsibilities:

- Developed reusable objects like PL/SQL program units and libraries, database procedures and functions, database triggers to be used by the team and satisfying the business rules.
- Used SQLServer Integrations Services (SSIS) for extraction, transformation, and loading data into target system from multiple sources
- Worked on development of data ingestion pipelines using ETL tool, Talend& bash scripting with big data technologies including but not limited to Hive, Impala, Spark, Kafka, and Talend.
- Experience in developing scalable& secure data pipelines for large datasets.
- Gathered requirements for ingestion of new data sources including life cycle, data quality check, transformations, and metadata enrichment.
- Built S3 buckets and managed policies for S3 buckets and used S3 bucket and Glacier for storage and backup on AWS.
- Developed Spark scripts to import large files from Amazon S3 buckets.
- Developed shell scripts for running Hive scripts in Hive and Impala.
- Used Jira for bug tracking and Bit bucket to check-in and checkout code changes
- Using Nebula Metadata, registered Business and Technical Datasets for corresponding SQL scripts
- Developed spark code and spark-SQL/streaming for faster testing and processing of data.
- Closely involved in scheduling Daily, Monthly jobs with Precondition/Post condition based on the requirement.

Environment: Scala, HDFS, AWS, Yarn, Hive, Sqoop, Flume, Kafka, Impala, Spark SQL, Spark Streaming, Eclipse, Oracle, Teradata, PL/SQL UNIX Shell Scripting.

Client: Avon Technologies Hyderabad India

July 2013 - September 2016

Role: Data Engineer

Responsibilities:

- Evaluated the traffic and performance of Daily deals PLA ads and compare those items with non-daily deal items to see the possibility of increasing ROI.
- Suggested improvements and modify existing BI components (Reports, Stored Procedures)
- Prepared Test Plan to ensure QA and Development phases are in parallel
- Developed PL/SQL procedures for different use case scenarios
- Involvement in post-production support, Testing and used JUNIT for unit testing of the module.
- Experienced in developing the UNIX Shell Scripts and PERL Scripts to execute the scripts and manipulate files and directory.html
- Experienced in working with spark ecosystem using SparkSQL and Scala queries on different formats like textfile, CSVfile.
- Have good working background with J2EE Frameworks like Servlets, JSP.
- Responsible for building scalable distributed data solutions using Data tax Cassandra.
- Implemented DAO layer using IBATIS and wrote queries for persisting demand core banking related information from the backend system using Query tool.
- Used Ajax to communicate with the server to get the asynchronous response.
- Developed code for Web services using XML, SOAP and used SOAP UI tool for testing the services.
- Developed SOAP and WSDL files for Web Services interacting with business Logic.
- Worked on to retrieve the data from FS to S3 using spark commands
- Using Nebula Metadata, registered Business and Technical Datasets for corresponding SQL scripts

Environment: Java/J2EE, JSP, XML, Spring Framework, Hibernate, Eclipse (IDE), Java Script, Ant, SQL, PL/SQL, Oracle, Windows, UNIX, Soap, Jasper reports.